Solution

Given lines $3x-4y-13=0$ ... (1)

$8x-11y=33$ ... (2)

$2x-3y-7=0$ ... (3)

Conditon for concurrent lines is $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$

i.e., $\begin{vmatrix} 3 & -4 & -13 \\ 8 & -11 & -33 \\ 2 & -3 & -7 \end{vmatrix} =$

$3(77-99)+4(-56+66)-13(-24+22)$

$=-66+40+26=0$

$\Rightarrow$ Given lines are concurrent.

To get the point of concurrency solve the equations (1) and (3)

Equation (1) $\times 2$ $\Rightarrow$ $6x-8y=26$

Equation (3) $\times 3$ $\Rightarrow$ $6x-9y=21$

$y=5$

When $y=5$ from (2) $8x=88$

$x=11$

Point of concurrency is (11, 5)

Example 3.8

If the lines $3x-5y-11=0$, $5x+3y-7=0$ and $x+ky=0$ are concurrent, find the value of k.

Solution

Given the lines are concurrent.

Therefore $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$

$\begin{vmatrix} 3 & -5 & -11 \\ 5 & 3 & -7 \\ 1 & k & 0 \end{vmatrix}=0$

$1(35+33)-k(-21+55)=0$

$\Rightarrow$ $34k=68$. $\therefore k=2$.

Example 3.9

A private company appointed a clerk in the year 2012, his salary was fixed as ₹20,000. In 2017 his salary raised to ₹25,000.

(i) Express the above information as a linear function in x and y where y represent the salary of the clerk and x-represent the year

(ii) What will be his salary in 2020?

Solution

Let y represent the salary (in Rs) and x represent the year

year (x)	salary (y)
2012(x_1)	20,000(y_1)
2017(x_2)	25,000(y_2)
2020	?

The equation of straight line expressing the given information as a linear equation in x and y is

$$\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}$$

$$\frac{y-20,000}{25,000-20,000}=\frac{x-2012}{2017-2012}$$

$$\frac{y-20,000}{5000}=\frac{x-2012}{5}$$

$y=1000x-2012000+20,000$

$y=1000x-19,92,000$

In 2020 his salary will be

$y=1000(2020)-19,92,000$

$y=$ ₹28000

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